

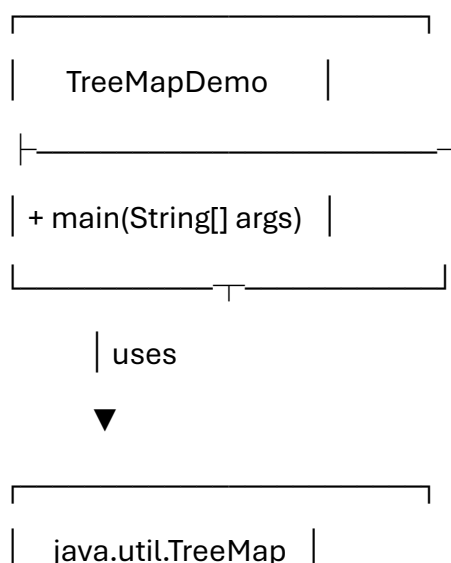
Aim

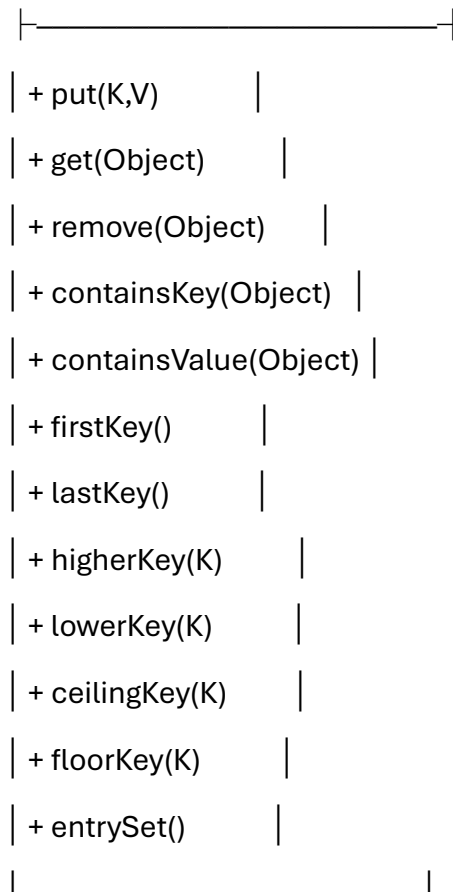
To write a Java program to demonstrate the important methods of the **TreeMap class**, such as `put()`, `get()`, `remove()`, `containsKey()`, `containsValue()`, `firstKey()`, `lastKey()`, `higherKey()`, `lowerKey()`, `ceilingKey()`, `floorKey()`, and `entrySet()`.

Algorithm

1. Start the program.
2. Import the `java.util.TreeMap` class.
3. Create a `TreeMap` object.
4. Add key-value pairs using `put()`.
5. Display the map in sorted key order.
6. Retrieve a value using `get()`.
7. Check for a key and value using `containsKey()` and `containsValue()`.
8. Remove a mapping using `remove()`.
9. Find the lowest and highest keys using `firstKey()` and `lastKey()`.
10. Find the keys immediately greater and smaller using `higherKey()` and `lowerKey()`.
11. Find the keys greater than/equal and smaller than/equal using `ceilingKey()` and `floorKey()`.
12. Display key-value pairs using `entrySet()`.
13. Stop the program.

UML Diagram





Code:

```

package Observations;

import java.util.TreeMap;

public class TreeMapClass {

    public static void main(String[] args) {

        // Creating TreeMap

        TreeMap<Integer, String> map = new TreeMap<>();

        // put()

        map.put(10, "Apple");

        map.put(20, "Banana");

        map.put(30, "Mango");
  
```

```
map.put(40, "Orange");
```

```
map.put(50, "Grapes");
```

```
System.out.println("TreeMap: " + map);
```

```
// get()
```

```
System.out.println("Value for key 30: " + map.get(30));
```

```
// containsKey()
```

```
System.out.println("Contains key 20: "
```

```
    + map.containsKey(20));
```

```
// containsValue()
```

```
System.out.println("Contains value Mango: "
```

```
    + map.containsValue("Mango"));
```

```
// remove()
```

```
map.remove(40);
```

```
System.out.println("After removing key 40: " + map);
```

```
// firstKey() and lastKey()
```

```
System.out.println("First key: " + map.firstKey());
```

```
System.out.println("Last key: " + map.lastKey());
```

```
// higherKey() and lowerKey()
```

```
System.out.println("Higher key than 20: "
```

```
    + map.higherKey(20));
```

```
System.out.println("Lower key than 30: "
```

```

        + map.lowerKey(30));

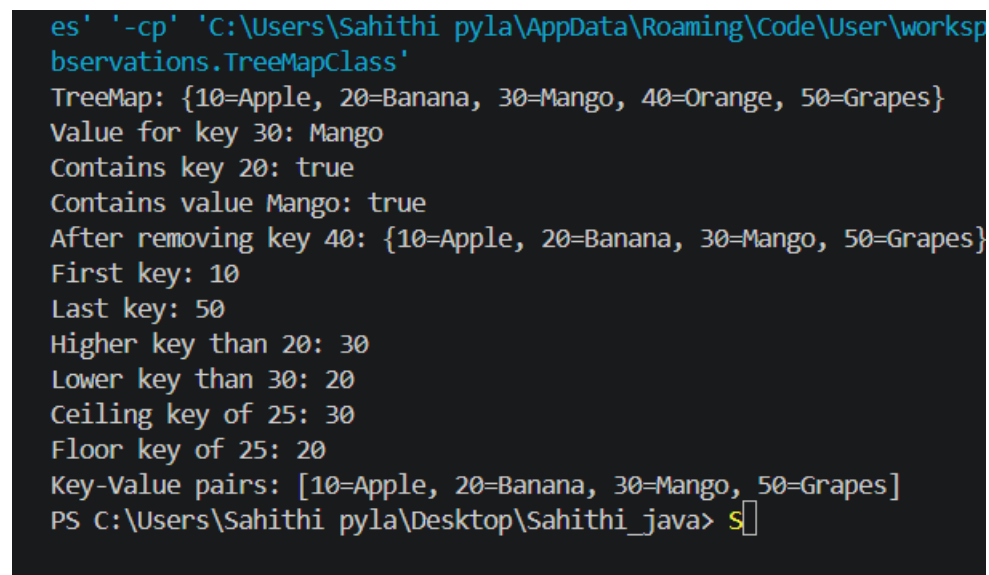
// ceilingKey() and floorKey()
System.out.println("Ceiling key of 25: "
        + map.ceilingKey(25));

System.out.println("Floor key of 25: "
        + map.floorKey(25));

// entrySet()
System.out.println("Key-Value pairs: "
        + map.entrySet());
}
}

```

Output:



```

es' '-cp' 'C:\Users\Sahithi\pyla\AppData\Roaming\Code\User\worksp
bservations.TreeMapClass'
TreeMap: {10=Apple, 20=Banana, 30=Mango, 40=Orange, 50=Grapes}
Value for key 30: Mango
Contains key 20: true
Contains value Mango: true
After removing key 40: {10=Apple, 20=Banana, 30=Mango, 50=Grapes}
First key: 10
Last key: 50
Higher key than 20: 30
Lower key than 30: 20
Ceiling key of 25: 30
Floor key of 25: 20
Key-Value pairs: [10=Apple, 20=Banana, 30=Mango, 50=Grapes]
PS C:\Users\Sahithi\pyla\Desktop\Sahithi_java> S

```

Result:

Thus, the Java program successfully demonstrates the important methods of the TreeMap class and shows that the key-value pairs are maintained in sorted order of keys.

